

# OBJECT ORIENTED ANALYSIS AND DESIGN

## UNIT – I

**Introduction to UML: Importance of modeling, principles of modeling, object oriented modeling, Conceptual model of the UML, Architecture, and Software Development Life Cycle.**

## UNIT – I

The UML is a graphical language for visualizing, specifying, constructing, and documenting the artifacts of a software-intensive system. The UML gives you a standard way to write a system's blueprints, covering conceptual things, such as business processes and system functions, as well as concrete things, such as classes written in a specific programming language, database schemas, and reusable software components.

### **Importance of modeling**

A model is a simplification of reality. A model provides the blueprints of a system. A model may be structural, emphasizing the organization of the system, or it may be behavioral, emphasizing the dynamics of the system.

### **Aims of modeling**

We build models so that we can better understand the system we are developing.

Through modeling, we achieve **four** aims.

1. Models help us to visualize a system as it is or as we want it to be.
2. Models permit us to specify the structure or behavior of a system.
3. Models give us a template that guides us in constructing a system.
4. Models document the decisions we have made.

We build models of complex systems because we cannot comprehend such a system in its entirety.

### **Principles of Modeling**

There are **four** basic principles of model

1. The choice of what models to create has a profound influence on how a problem is attacked and how a solution is shaped.
2. Every model may be expressed at different levels of precision.
3. The best models are connected to reality.
4. No single model is sufficient. Every nontrivial system is best approached through a small set of nearly independent models.

### **Object Oriented Modeling**

In software, there are several ways to approach a model. The **two** most common ways are

1. Algorithmic perspective
2. Object-oriented perspective

#### **Algorithmic Perspective**

- The traditional view of software development takes an algorithmic perspective.
- In this approach, the main building block of all software is the procedure or

## Conceptual model of UML

To understand the UML, you need to form a **conceptual model of the language**, and this requires learning three major elements:

- **Basic building blocks of the UML**
- **Rules**
- **Common Mechanisms in the UML**

**Basic building blocks of the UML:** Vocabulary of the UML can be defined

1. Things
2. Relationships
3. Diagrams

### Things in the UML

There are **four** kinds of things in the UML:

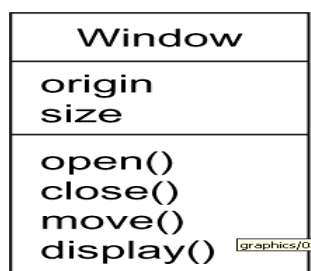
- Structural things
- Behavioral things
- Grouping things
- Annotational things

**Structural things** are the nouns of UML models. These are the mostly static parts of a model, representing elements that are either conceptual or physical. In all, there are **seven** kinds of structural things.

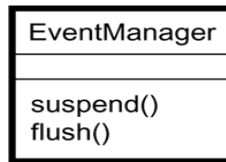
1. Classes
2. Interfaces
3. Collaborations
4. Use cases
5. Active classes
6. Components
7. Nodes

### Class

- It is a description of a set of objects that share the same attributes, operations, relationships, and semantics.
- A class implements one or more interfaces.
- Graphically, a class is rendered as a rectangle, usually including its name, attributes, and operations.



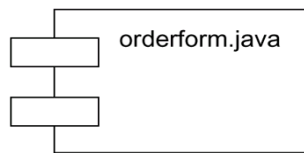
**Fig: Class**



**Fig: Active class**

### Component

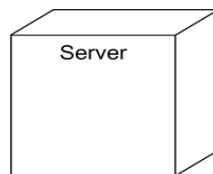
- It is a physical and replaceable part of a system that conforms to and provides the realization of a set of interfaces.
- Graphically, a component is rendered as a rectangle with tabs.



**Fig: Component**

### Node

- It is a physical element that exists at run time and represents a computational resource, generally having at least some memory and, often, processing capability.
- Graphically, a node is rendered as a cube, usually including only its name.



**Fig: Node**

### Behavioral Things

- They are the dynamic parts of UML models. These are the verbs of a model, representing behavior over time and space.
- In all, there are **two** primary kinds of behavioral things.

Interaction

State machine

### Interaction

- Interaction is a behavior that comprises a set of messages exchanged among a set of objects within a particular context to accomplish a specific purpose
- An interaction involves a number of other elements, including messages, action sequences and links
- Graphically a message is rendered as a directed line, almost always including the name of its operation



**Fig: Message**

### State Machine

- State machine is a behavior that specifies the sequences of states an object or an interaction goes through during its lifetime in response to events, together with its responses to those events